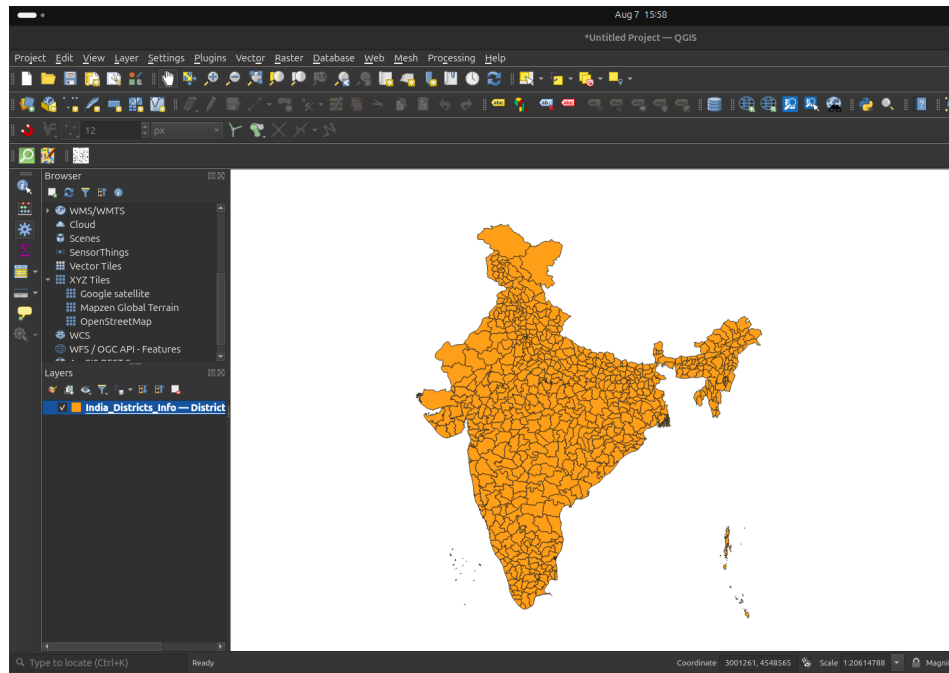


Thematic Mapping Manual

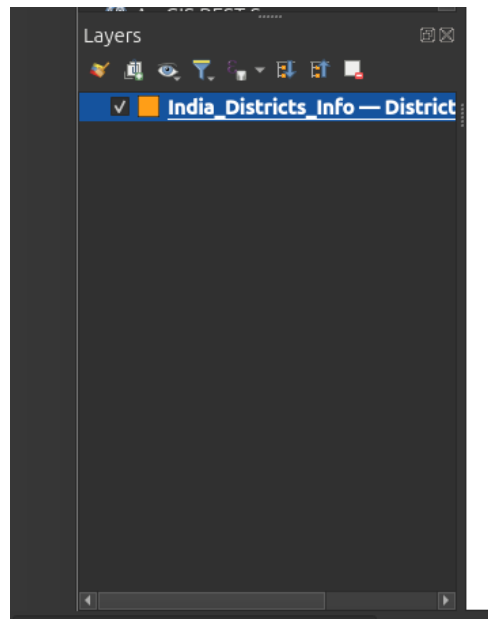


1 Add the Layer

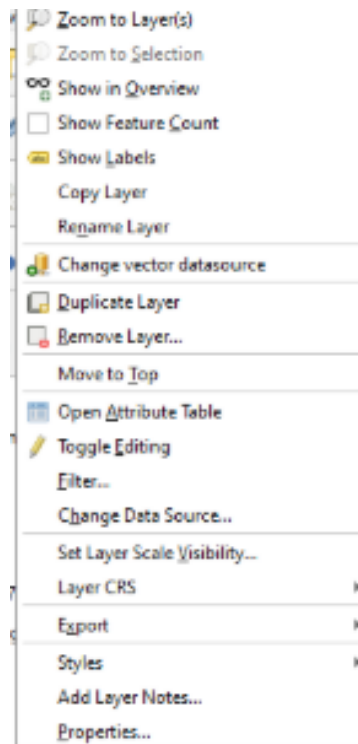
We will create a choropleth map, a thematic mapping technique in which geographic regions are symbolized with different colours or shades according to the values of a selected attribute. It is used to visualize the spatial distribution and variation of attribute data across geographic areas.

2 Step-by-Step Instructions for Creating a Choropleth Map

1. Right-click the Districts layer in the Layers panel on the left.

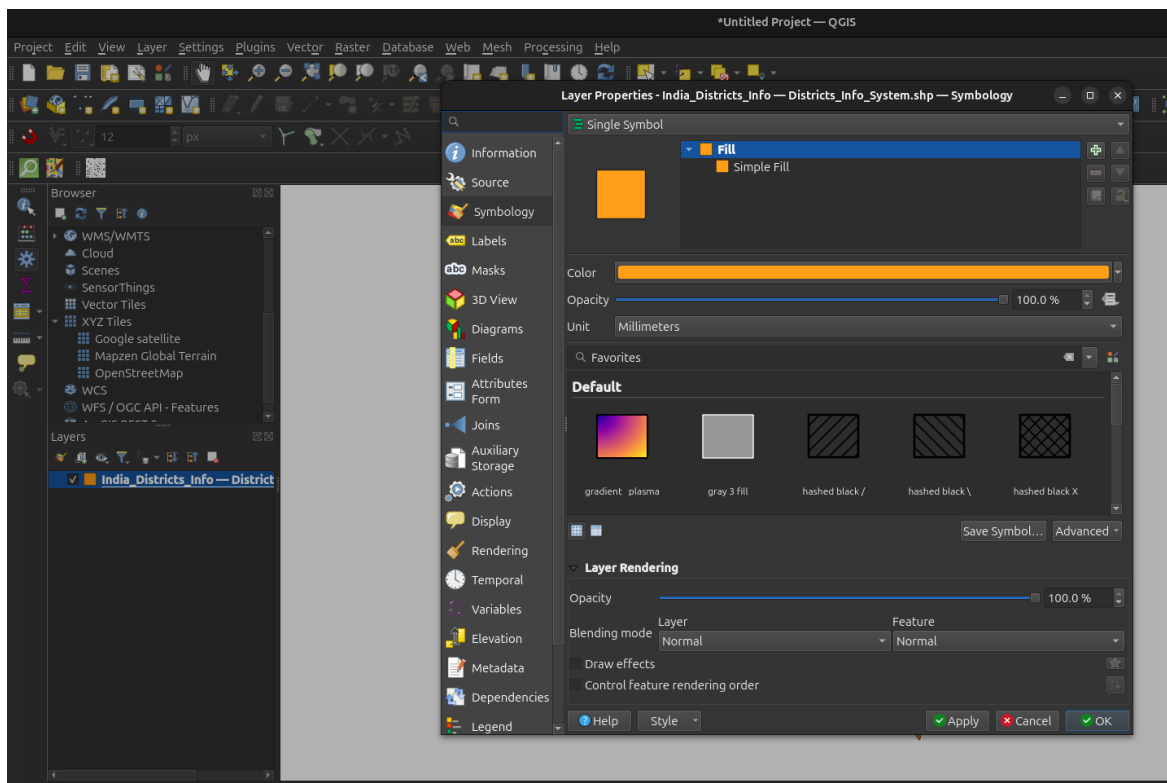


2. You will see options like these.

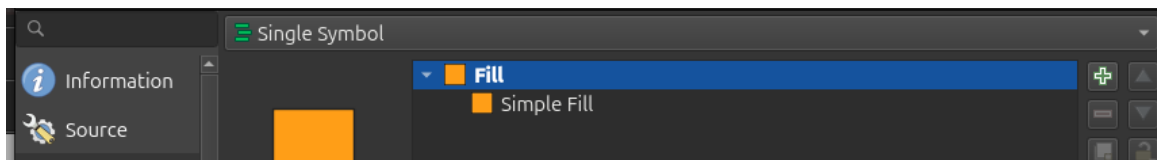


3. Click **Properties**, which is at the bottom of the list.

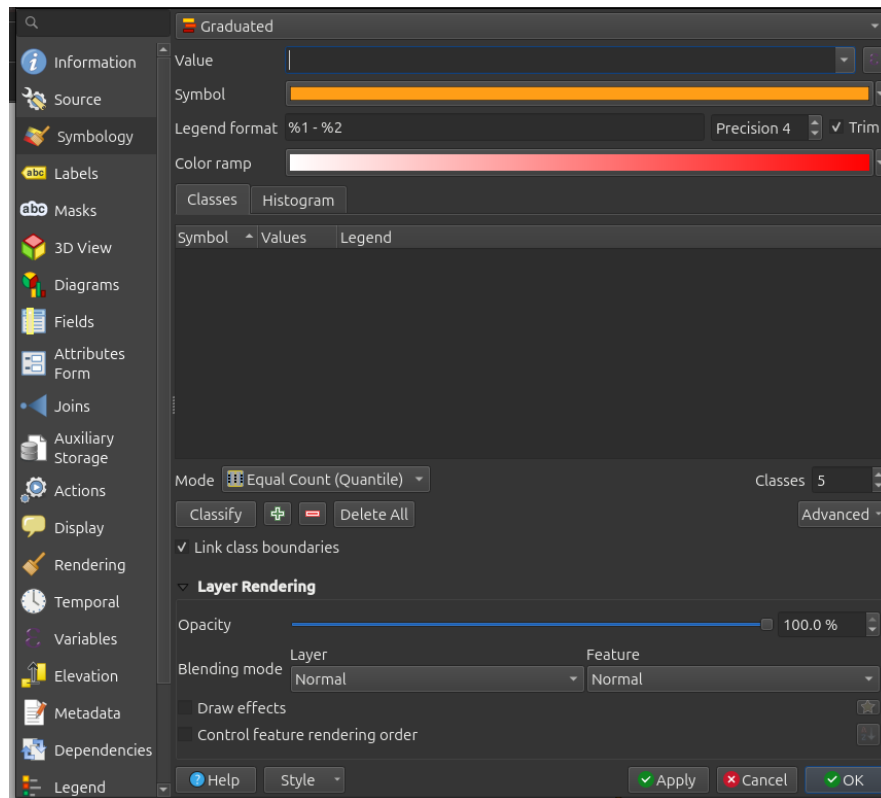
4. Once you click **Properties**, you will see a screen like this.



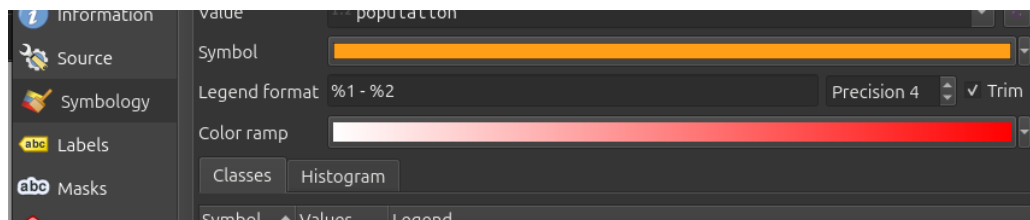
5. Here, click **Single Symbol**, which is at the top of the page.



6. You will see a drop-down menu containing methods for symbolizing features. Choose **Graduated** because we want to symbolize features using the values of a numeric attribute.

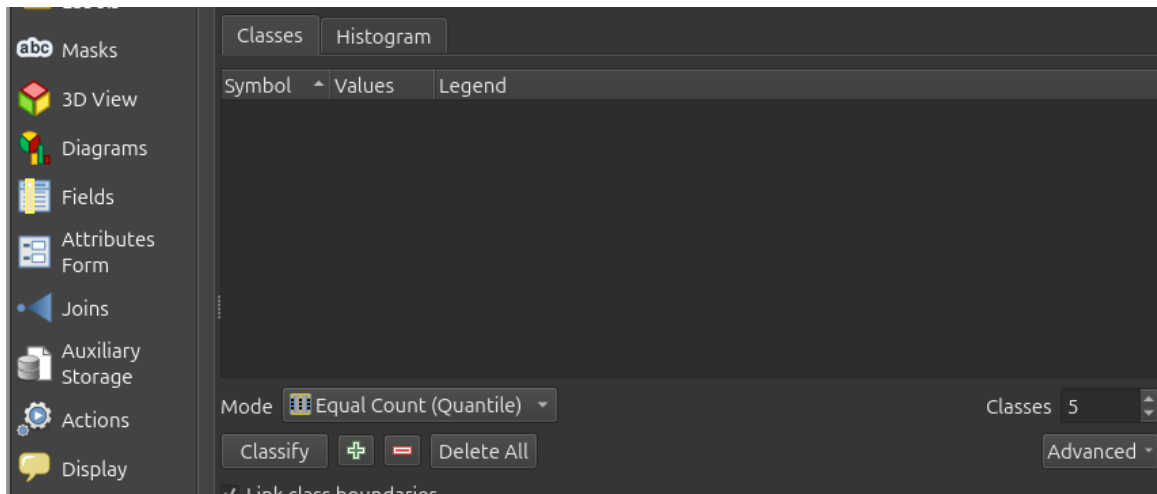


7. In the **Value** box, select the attribute field (column) whose values will be used to classify and colour the features. In this example, we selected **Population**; however, you can choose any other appropriate numeric attribute.

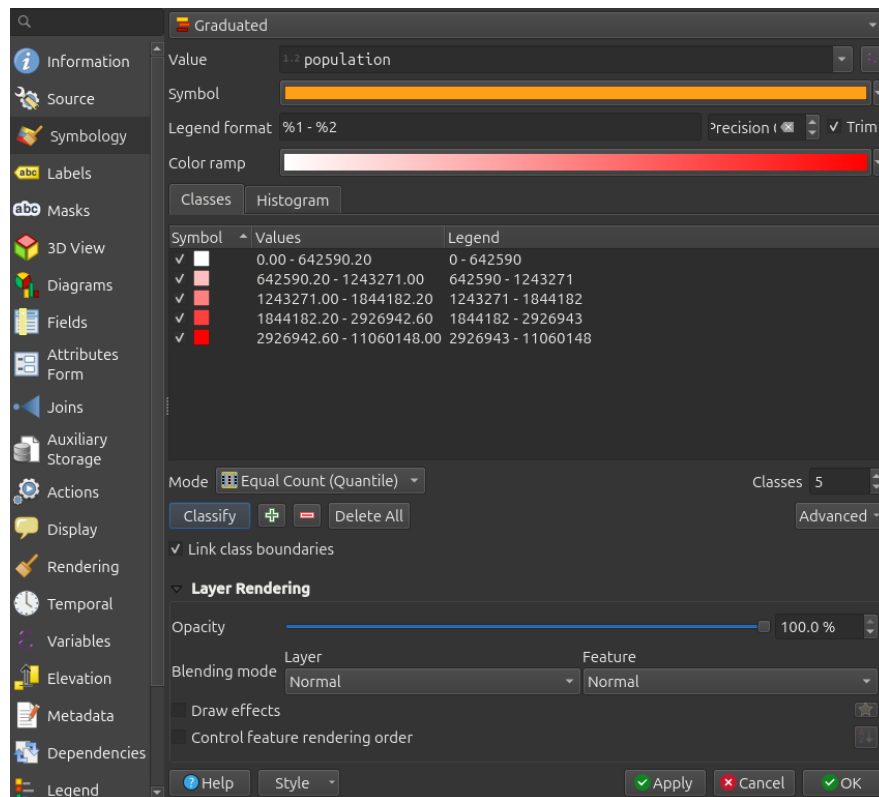


The **Symbol** defines the basic appearance of each feature before colours are applied.

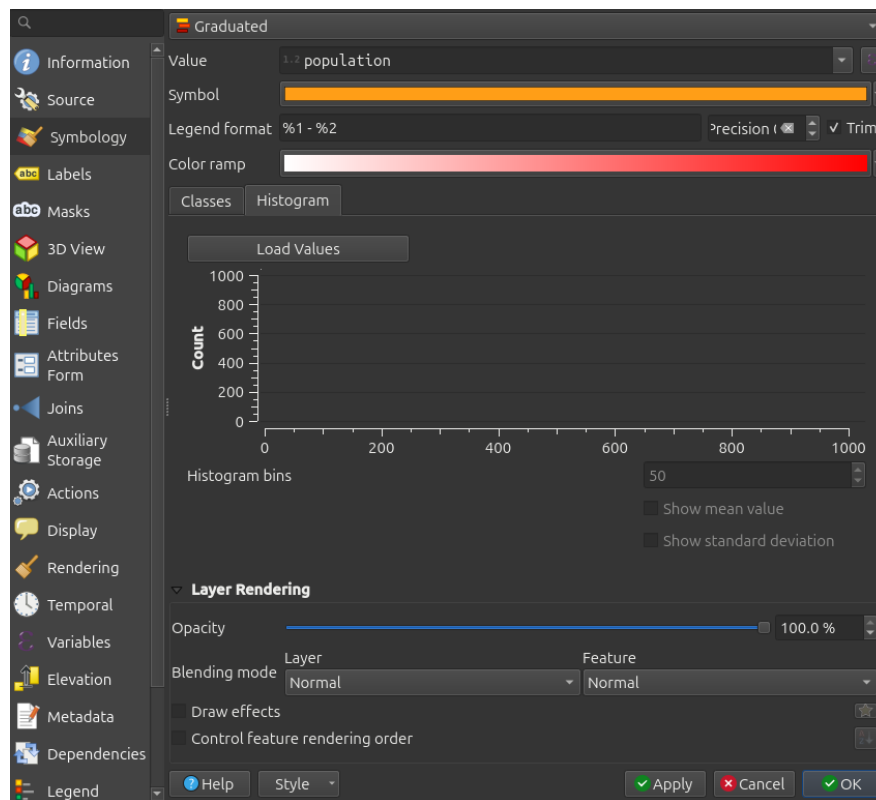
The **Color Ramp** determines which colours represent low and high attribute values.



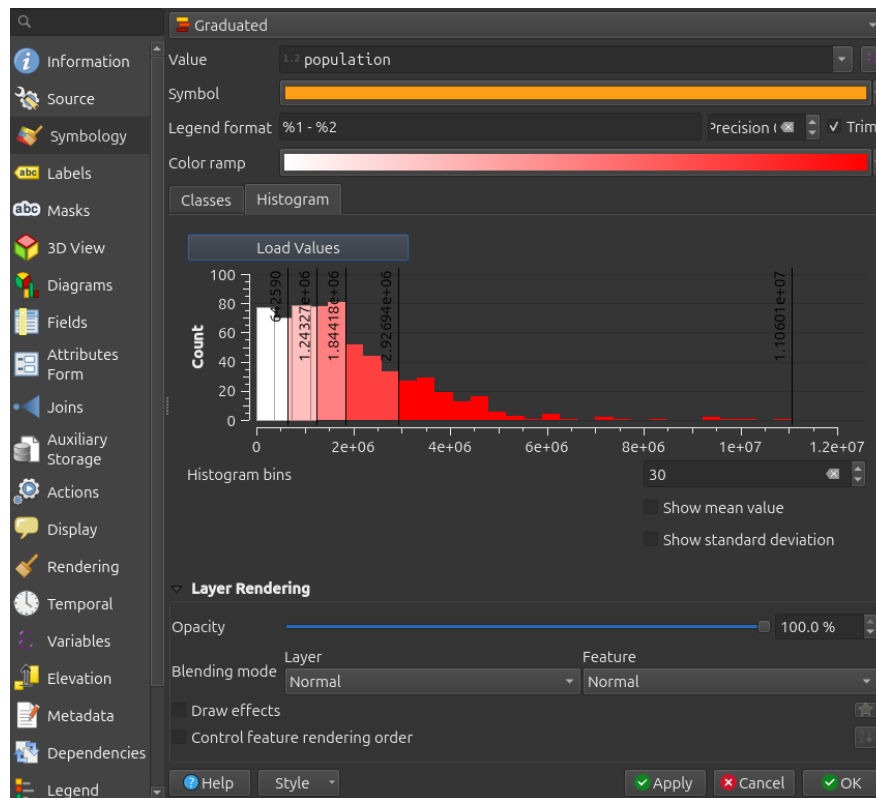
8. Below the **Color Ramp**, you will find the **Classes** and **Histogram** tabs. Use the **Mode** drop-down menu to select a classification method (most are discussed in class), and use the **Classes** field on the right to change the number of classes.
9. After selecting the mode and number of classes, click **Classify**, which is located just below **Mode**.



10. Once you click **Classify**, you will see the five classes. To view the histogram, click the **Histogram** tab beside the **Classes** tab.

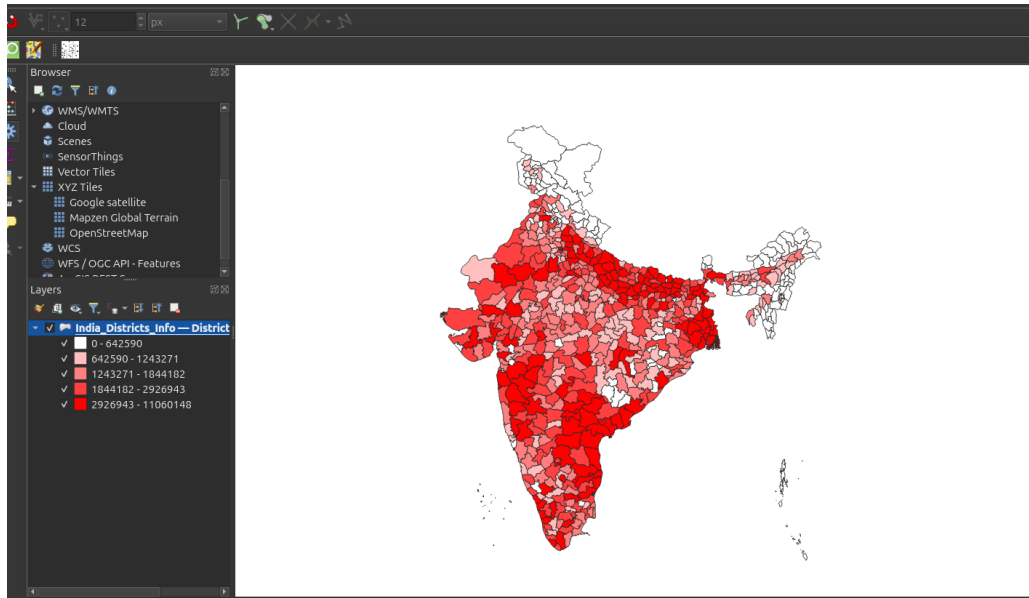


11. Click **Load Values**.



You can now see the histogram, which shows the distribution of the selected attribute—in this example, **Population**.

12. Click **Apply**, then click **OK**.



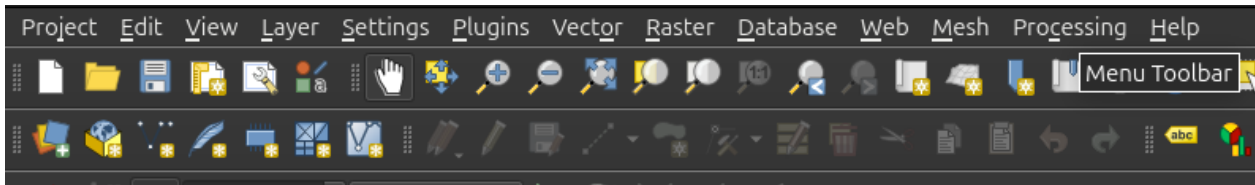
You will see a map like this. Your map may vary depending on the data and classification settings you choose.

3 Graduated and Proportional Symbol Mapping

For a Graduated and Proportional symbol map, we will first create a centroid for each district.

3.1 Open the Processing Toolbox

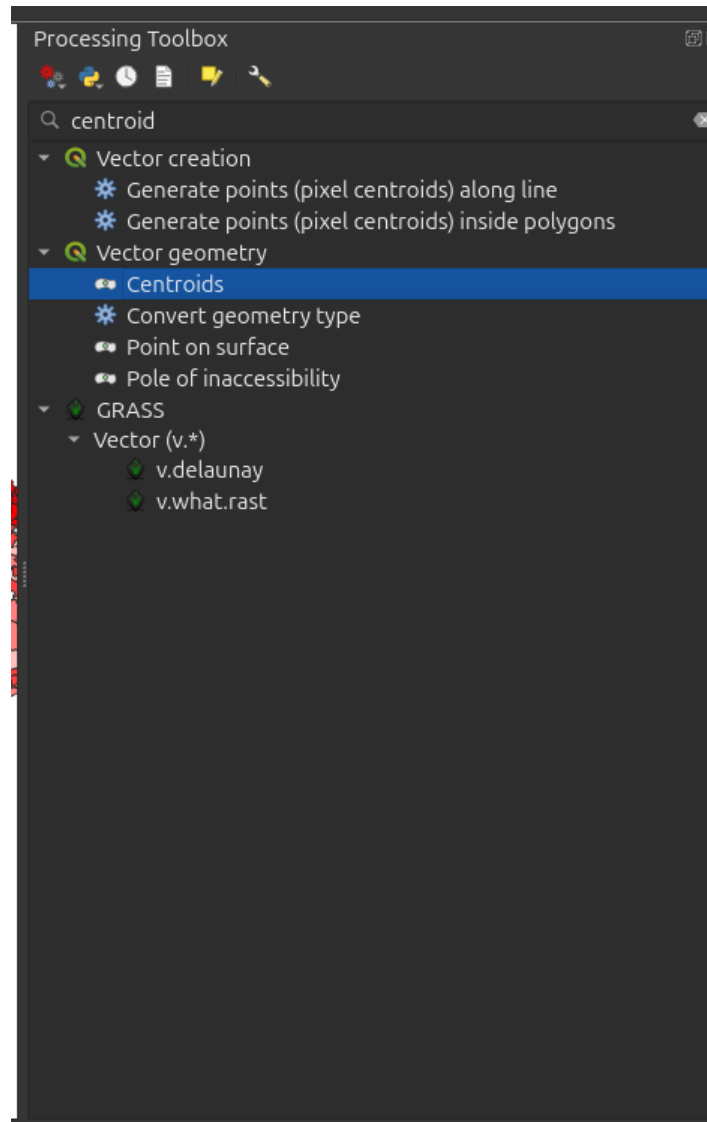
1. Go to the menu bar at the top of the page, click **Processing**, and then select **Toolbox**.



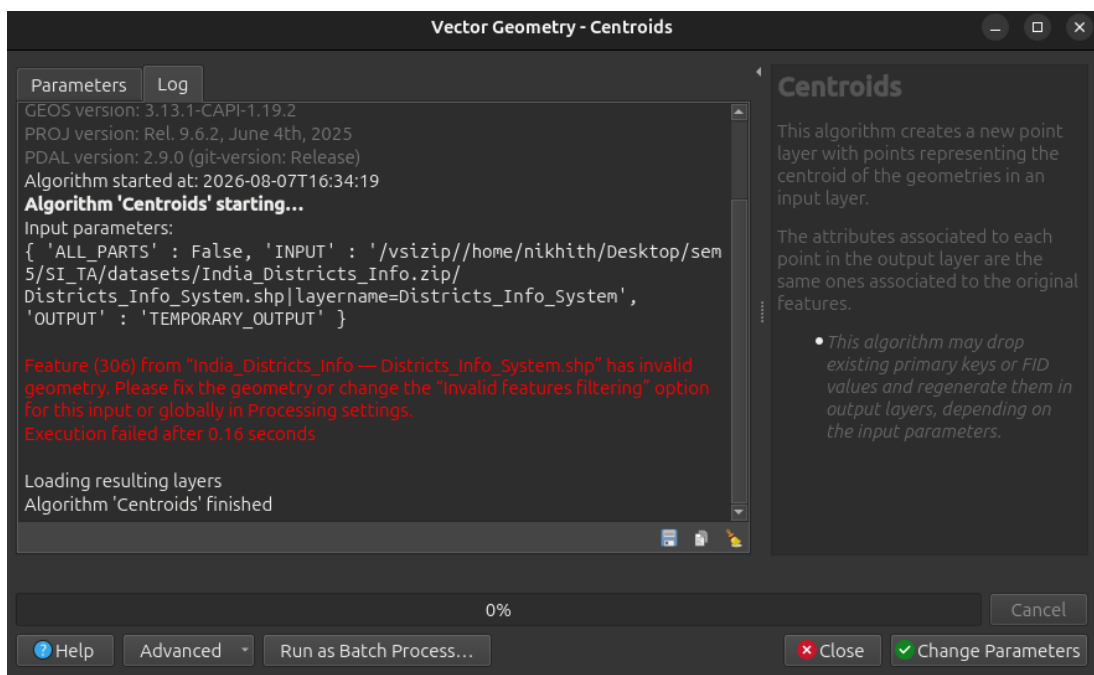
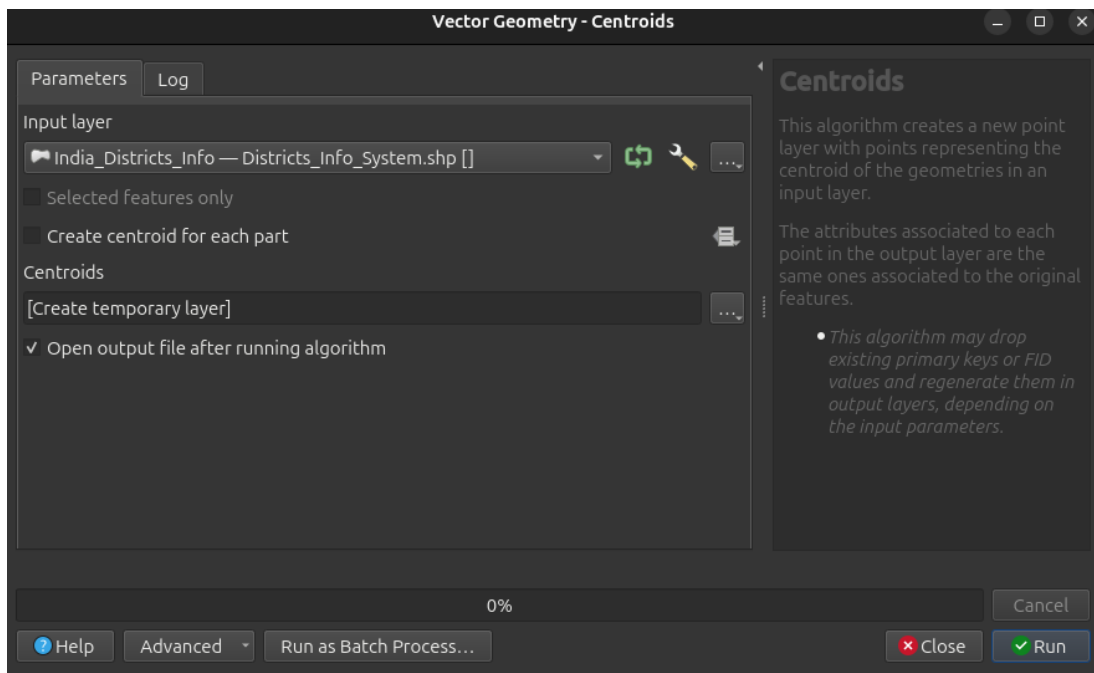
2. The **Processing Toolbox** will appear on the right side of the page.



3. In the search box, type **Centroids**, then click **Centroids** in the results.



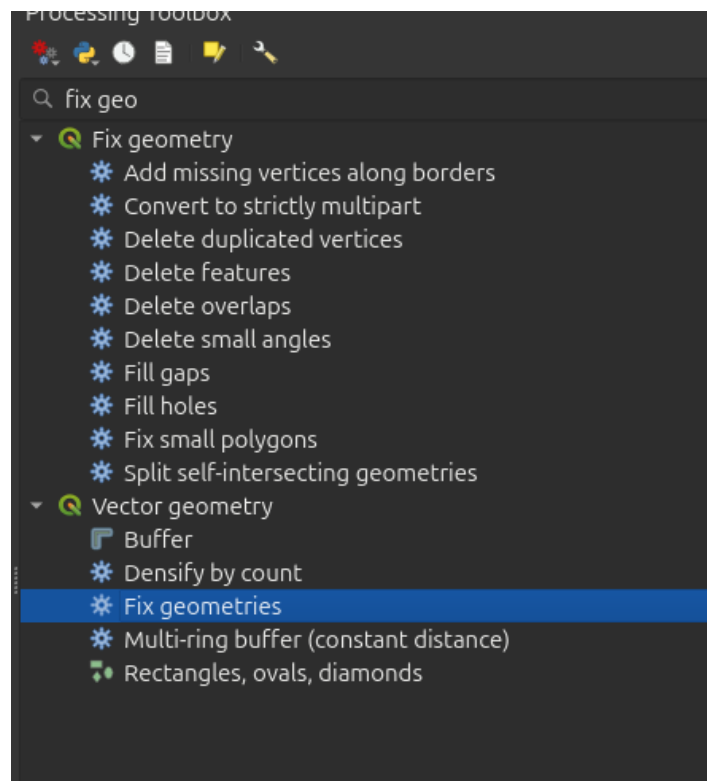
4. You will see the Centroids tool dialog. Select the Districts layer as the input layer, then click **Run**.



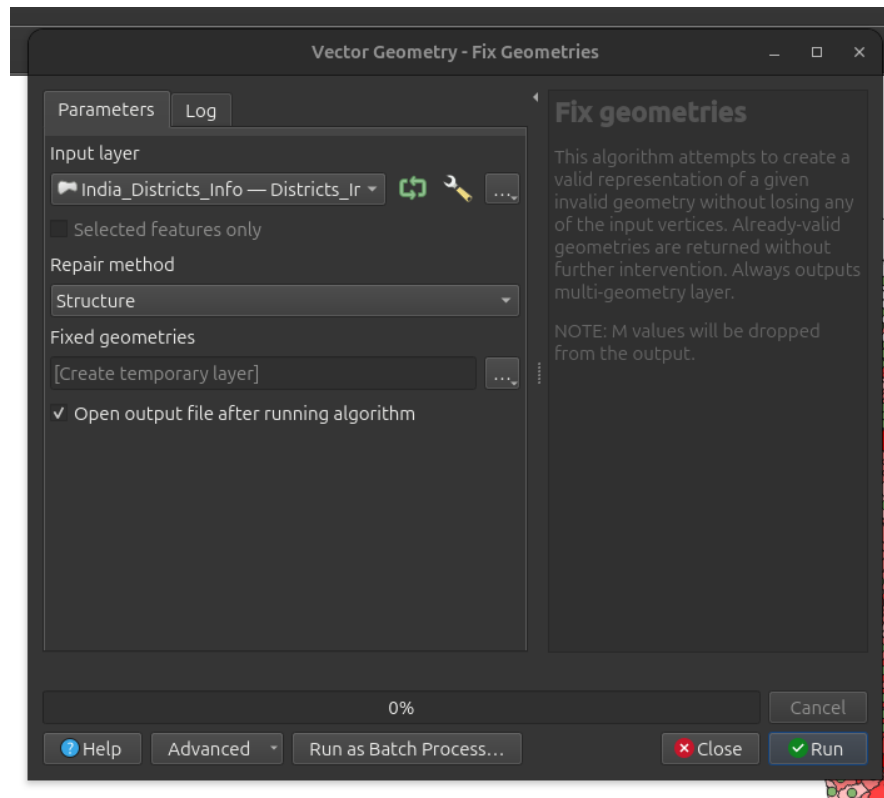
3.2 Fix Invalid Geometries

If you receive an error like the one shown above, use the **Fix Geometries** tool before running the Centroids tool again.

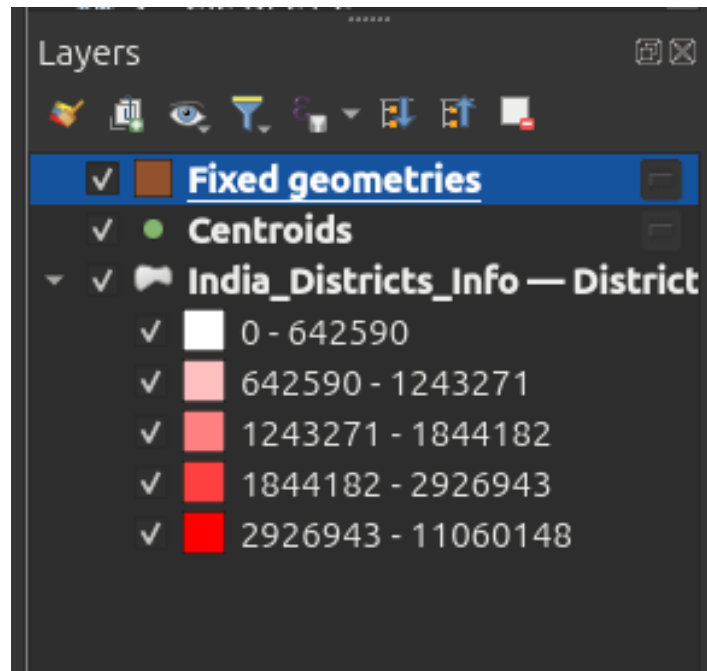
1. In the Processing Toolbox search box, search for **Fix Geometries**, then click it in the results.



2. Select the Districts layer as the input layer, then click **Run**.



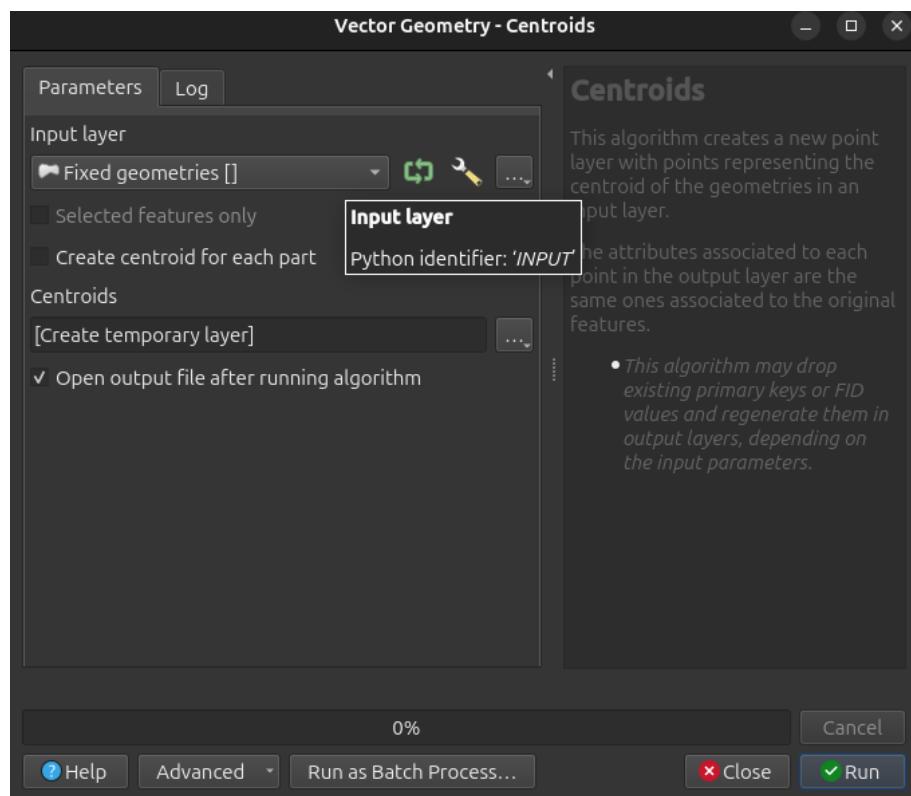
3. After the tool finishes, a layer named **Fixed geometries** will appear in the Layers panel.



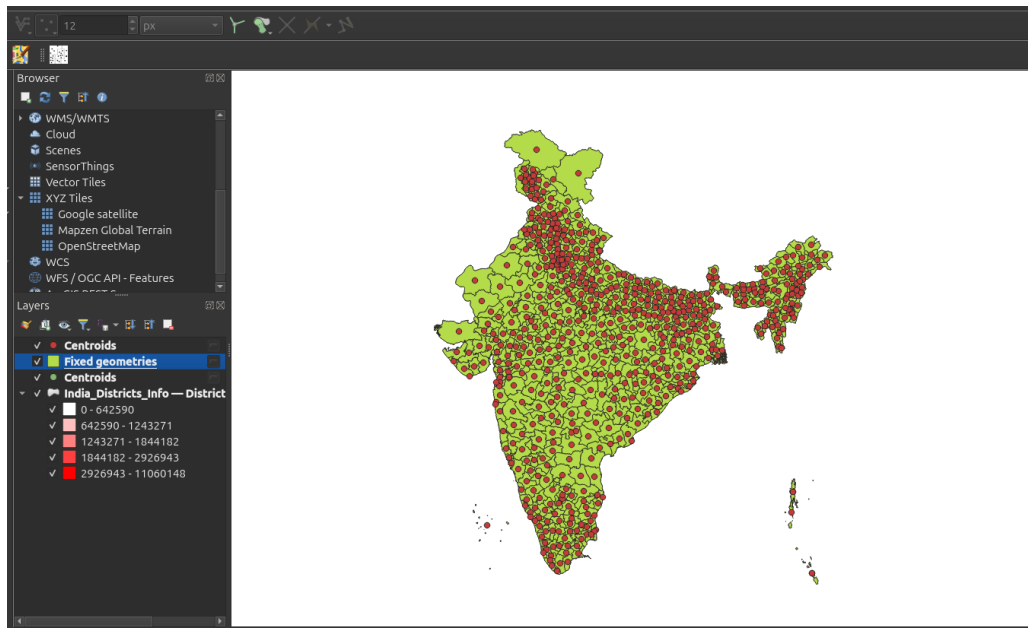
3.3 Create Centroids from the Fixed Layer

Repeat the Centroids procedure described above, but select **Fixed geometries** as the input layer.

1. In the Centroids tool dialog, choose **Fixed geometries** as the input layer, then click **Run**.



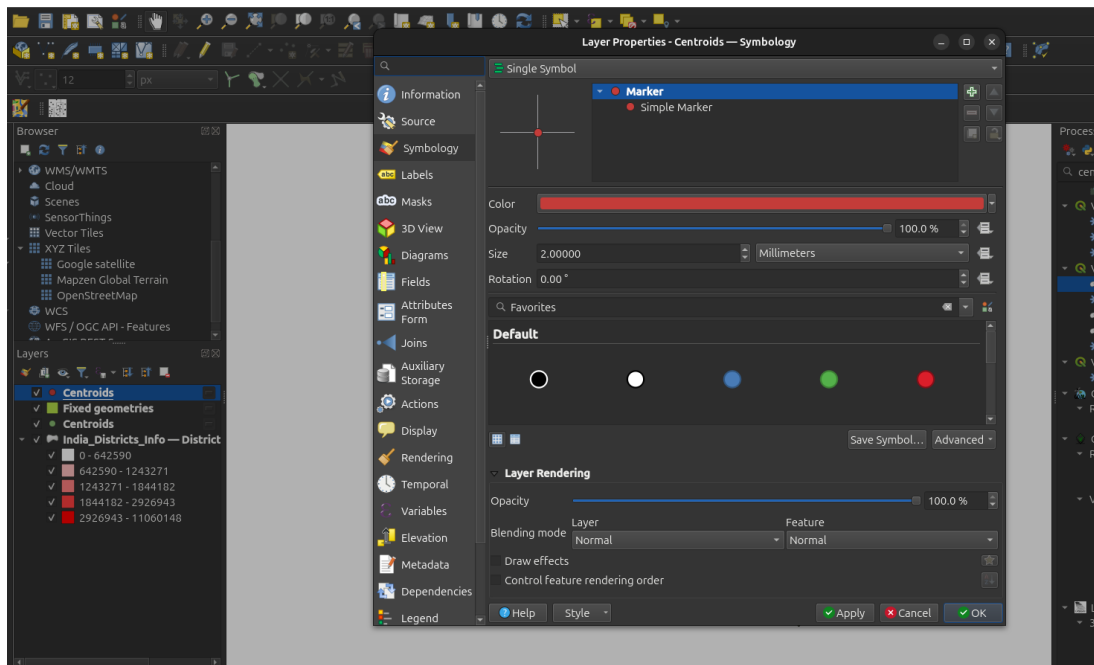
2. After the tool finishes, the Centroids layer will be displayed on the map.



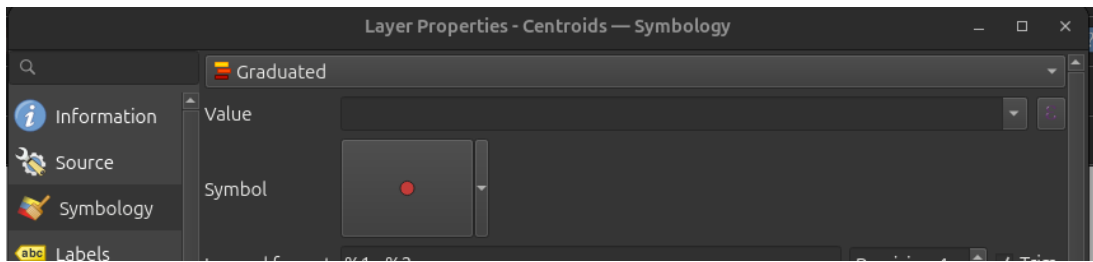
3.4 Style the Centroids as Graduated Symbols

The Centroids layer now displays one point for each district.

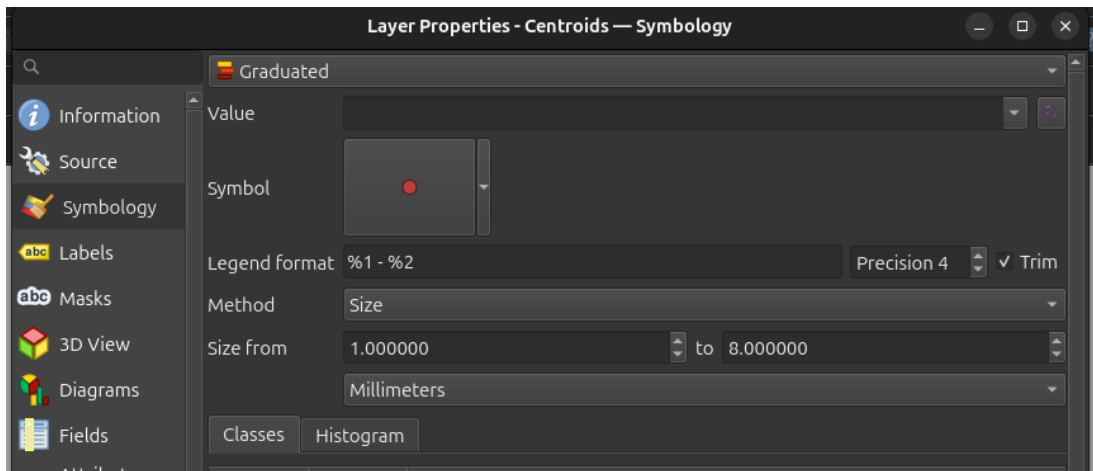
1. Right-click the **Centroids** layer and select **Properties**.



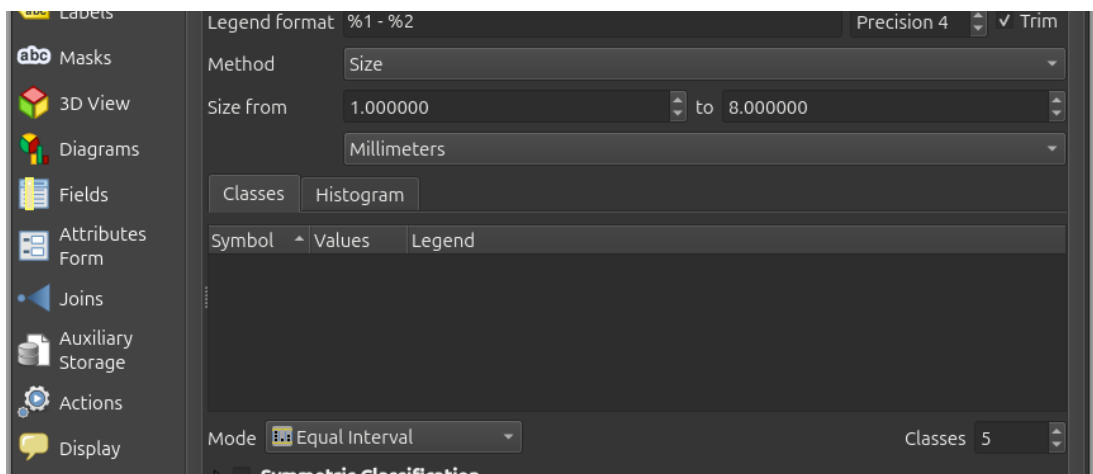
2. In the Symbology tab, click the **Single Symbol** drop-down menu at the top and select **Graduated**.



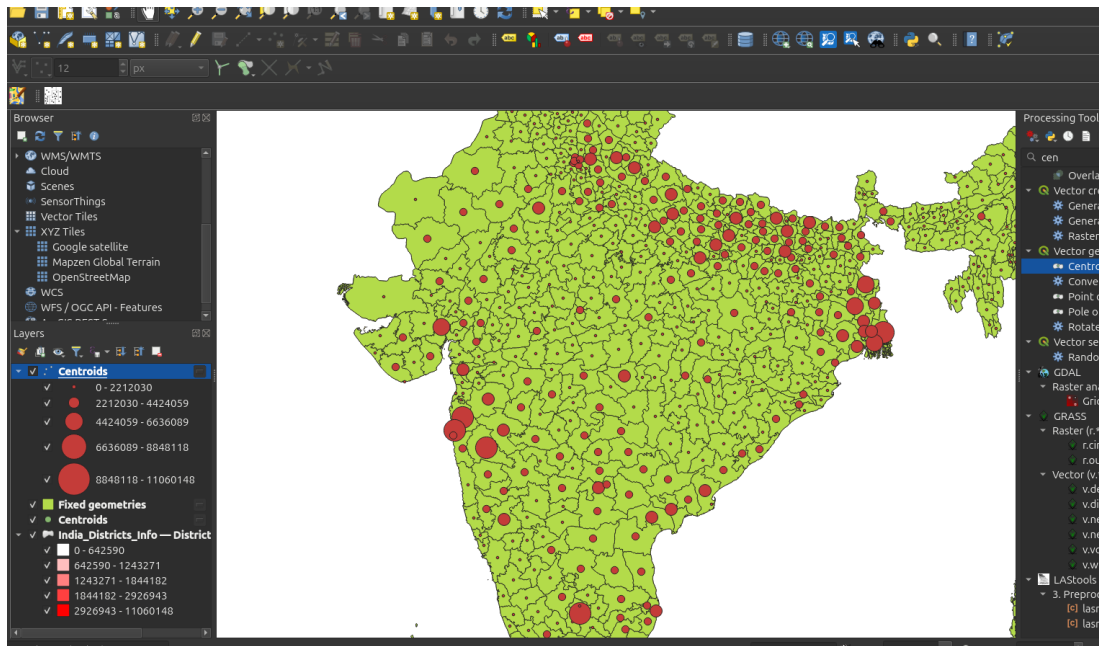
3. In the **Value** box, select the attribute field (column) whose values will be used to classify the features. Then change the default **Method** from **Color** to **Size**.



4. As with the Districts layer, choose a classification mode from the **Mode** drop-down menu. In this example, we selected **Equal Interval**.



5. Click **Apply**, then click **OK**.

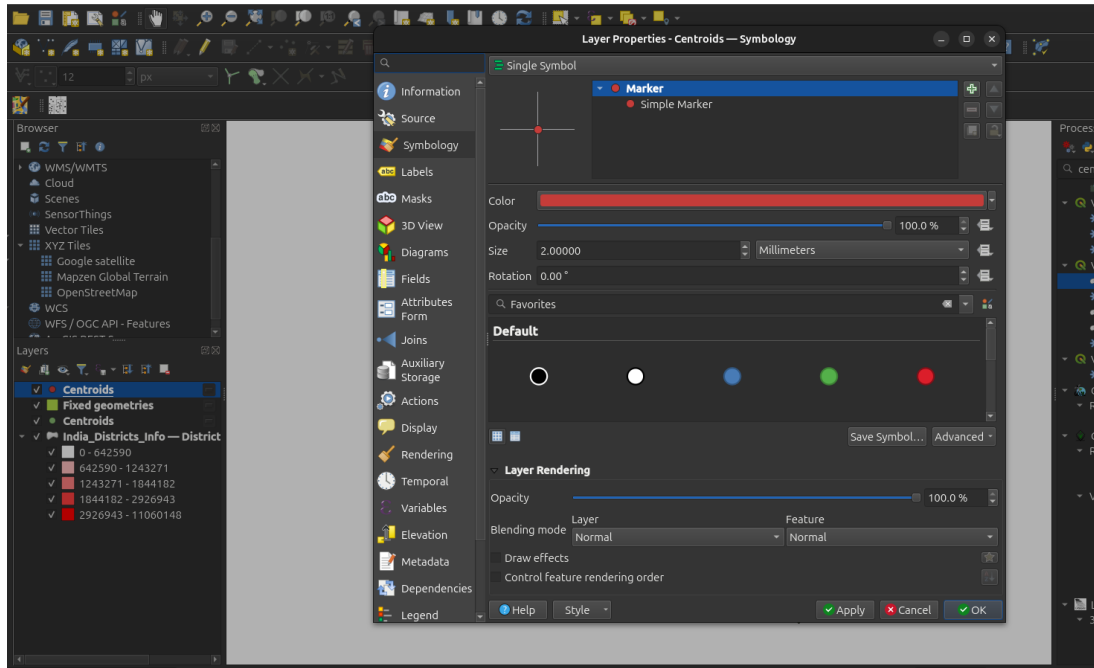


You will see a map like this. The circle size represents the population.

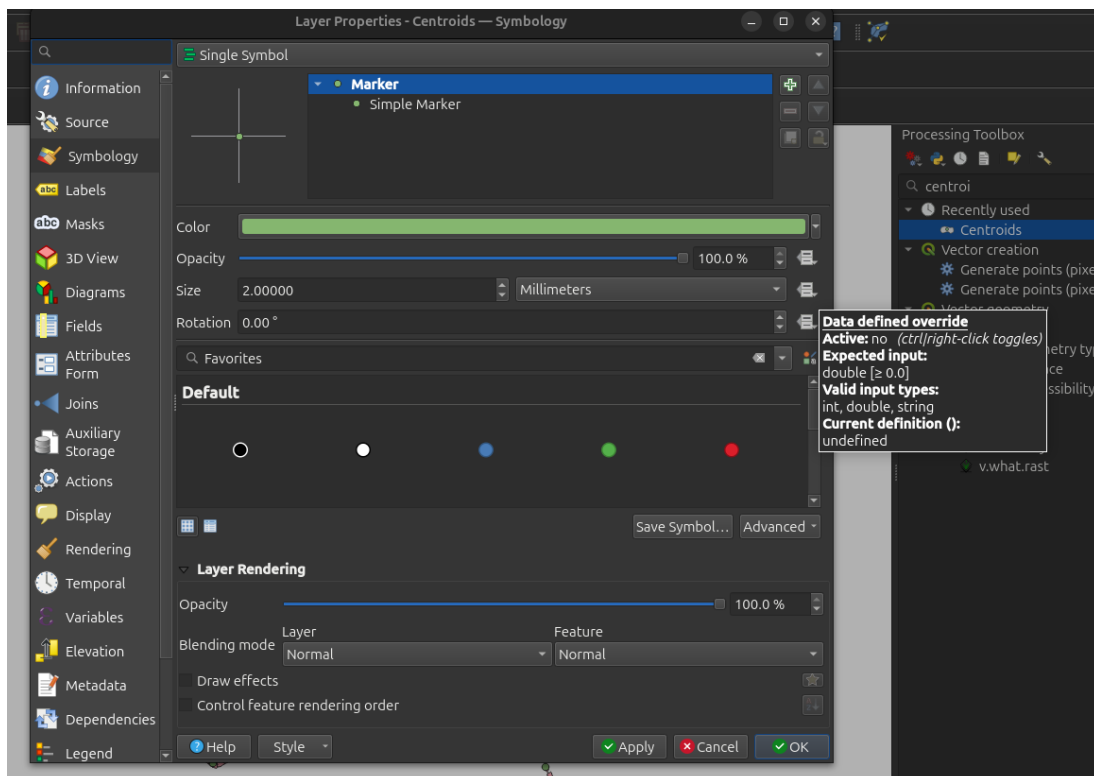
3.5 Style the Centroids as Proportional Symbols

The Centroids layer now displays one point for each district.

1. Right-click the **Centroids** layer and select **Properties**.



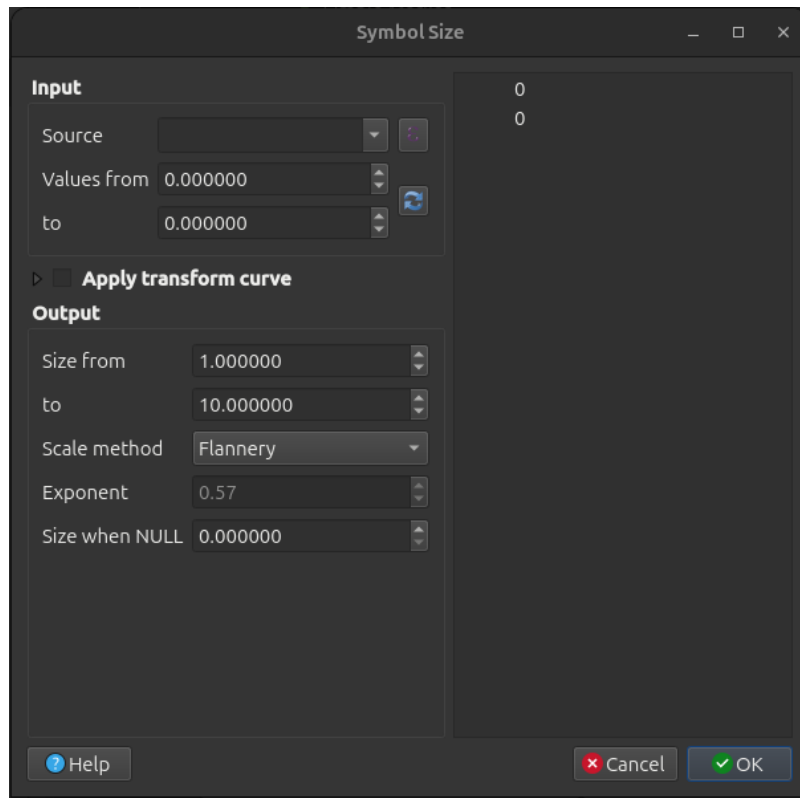
2. In the Symbology tab, use the **Single Symbol** settings shown below.



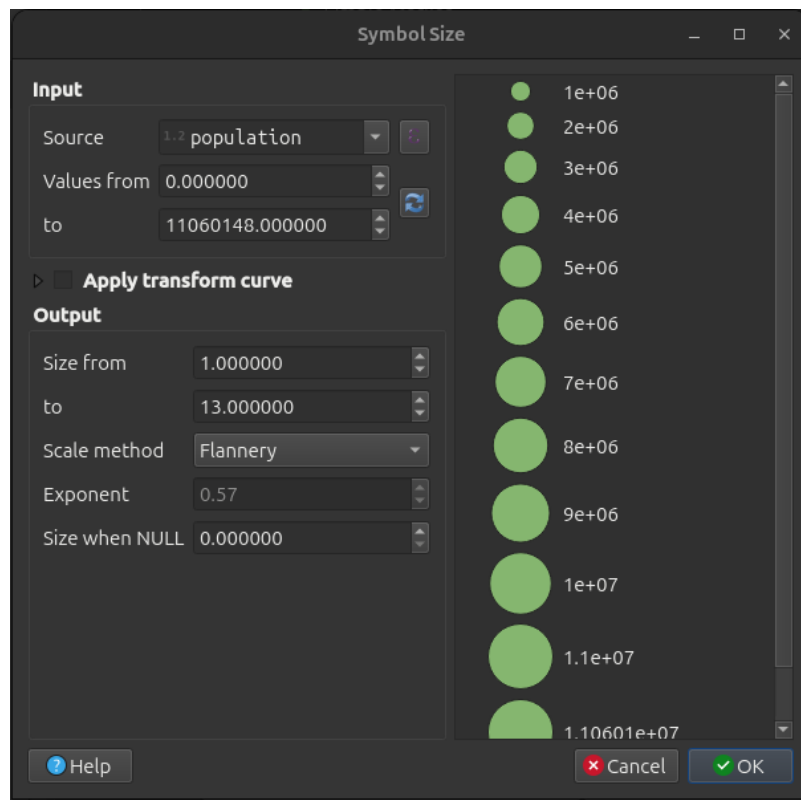
3. Click the **data-defined override** button just beside the **Size** option.



4. The data-defined override menu will open. Click **Assistant**, which is at the bottom of the list. You will then see the Symbol Size dialog shown below.



5. In **Source**, select the attribute you want to use, then click **OK**. In this example, we selected **Population**.



6. The map will now look like this, with symbol sizes representing the selected attribute values.

